MCA -I Sem – I

Division – D

Name: Rushikesh Ganesh Yadav

Roll No. : MC2527261

Subject: [IT11L] Python Programming Lab

For Academic Year 2025 – 2026

Department of Master of Computer Application

CERTIFICATE

This is to certify that, **Mr. Rushikesh Ganesh Yadav** of **MCA-D (SEM-I)** Class, Roll No. **MC2527261**, has completed all the Practical work of the subject

"IT11L Python Programming Practical Assignments" satisfactorily in the MCA program as prescribed by Savitribai Phule Pune University in the academic year

2025 - 2026.

Date :

Course Coordinator **HOD, MCA** **Director**

Examiner 1 Sign **Examiner 2 Sign**

INDEX

Sr. No.	Practical Name	Date	Sign
1.	Write a python program to print the following string in a specific format? Out put : Twinkle, twinkle, little star, How I wonder what you are! Up above the world so high, Like a diamond in the sky. Twinkle, twinkle, little star, How I wonder what you are		
2	Write a program to perform arithmetic operations.		
3	Write a program using list to perform below operation: a. Slicing b. Concatenation c. Repetition		
4	Write a program using tuple to perform below operation: a. Slicing b. Concatenation c. Repetition		
5	Write a program using dictionary to access value using keys .		
6.	Write a program using set: a. Create empty set b. Assign value to set c. Adding element d. Removing element		
7.	. Write a program to get the largest number from the given list ?		
8.	Write a program to find out Factorial of a given Number		
9.	Write a program to find given number is prime or not		
10.	Write a program to identify total number of digits present in given number		

11.	Write a program to display Fibonacci series up to a given number by using range function		
12.	Write a program using for loop with: String, Integer, Enumerate, Nested, For List, Dictionary, Tuple, Zip()		
13.	Write a program to perform Loop manipulation using pass, continue, break and else		
14.	Write a program to perform Comprehensions on List Dictionaries.		
15.	Write a program to Find all the numbers from 1-500 that have three in them.		
16..	Write a python program for numbers=range(20) produce a list containing the word "even" if the number is even or the word "odd" if the number is odd		
17	Find the common numbers in two list 1[1,3,4,5,6,7] 2[5,2,8,7,1,3]		

ASSIGNMENT=2

1	Implement a nested function that calculates compound interest using non-local variables		
2	Write a program using lambda to filter even numbers from a list		
3.	Write a program using lambda to find area of triangle		
4.	Create a generator function to yield Fibonacci numbers up to a certain limit		
5.	Develop a custom Python module with functions for basic statistical operations (mean, median, mode)		
6.	rite a program to handle file reading errors and log them to a separate file		
7	reate a custom exception class AgeException and use it to validate user input		

ASSIGNMENT=3

1	Write a simple Python class named Student and display its type. Also display the dict attribute keys and values of the module attribute of the Student class.		
2	Write a Python class Student with attributes, student name and marks. Modify any attribute value of the class and show the original and modified values o attributes.		
3	Write a Python class to convert an integer to a Roman numeral.		
4	Write a Python class to convert a Roman numeral to an integer		
5	W ite a program to create a class Shape with methods called get_perimeter and get area. Create a subclass called Circle that overrides the get_perimeter and get area methods o calculate the area and erimeter of a circle.		
6	rite a program to create a Vehicle class hierarchy. The base class should be ehicle, with subclasses Truck, Car, and Motorcycle. Each subclass should ave properties such as make, model, year, and fuel_type. Implement methods for calculatin fuel efficienc , distance traveled, and maximum s eed.		
7	Implement a class BankAccount with methods for deposit, withdrawal, and dis la in the account balance.		
8	Develop a hierarchy of shapes (e.g., Shape, Circle, Rectangle) with area and perimeter methods.		
9	Overload the + operator for concatenating two custom string objects		

10	Create a static method to validate user credentials (e.g., email format, password len h		
11	Implement a class Library containing a list of Book objects		
12	Demonstrate method overriding in a superclass-subclass relationship for a payment		
13	Write a program to validate a password that meets certain criteria (length, uppercase, dioit s ecial character .		
14	Parse email addresses from a given text and classify them by domain		
15	Create threads to perform independent tasks such as downloading multiple files Simultaneousl		
16	Write a Python program to create two threads. The first thread will print even numbers, and the second thread will print odd numbers. Demonstrate the conce t of multithreadin		
17	Use multithreading to compute factorials of multiple numbers and log the results into a file.		

ASSIGNMENT=4

1.	Develop a Python program to create, read, update, and delete documents in MongoDB a collection.		
2.	Write a program to insert valid data into MongoDB using exception handling for errors.		
3.	Use MongoDB's aggregation pipeline to find the average salary of employees in a collection.		
4.	Develop a Python program to create a database name employee. With the collection name empinfo. Add 5 douments in the collection. Perform read, update, and delete operations on documents in a MongoDB.		

ASSIGNMENT=5

1	Create a Django project with models for Product and Category, along with views to display them on a webpage.		
2.	mplement user login, logout, and registration functionality in a Django application.		
3.	Build a REST API for a task management system where users can create, update, and delete tasks		

UNIT-I

Python Practical Assignment-1

1. Write a python program to print the following string in a specific format?

Out put :

Twinkle, twin

How I wonder what you are!

Up above the world so high,

Like a diamond in the sky.

Twinkle, twinkle, little star,

How I wonder what you are

Ans-

```
print("Twinkle, twin")
print("How I wonder what you are!")
print("Up above the world so high,")
print("Like a diamond in the sky.")
print("Twinkle, twinkle, little star,")
print("How I wonder what you are")
```

2. Write a program to perform arithmetic operations.

Ans-

```
# Taking input
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
# Performing operations
print("Addition (+):", a + b)
print("Subtraction (-):", a - b)
print("Multiplication (*):", a * b)
print("Division (/):", a / b)
```

Output-

Enter first number: 10

Enter second number: 4

Addition (+): 14

Subtraction (-): 6

Multiplication (*): 40

Division (/): 2.5

3. Write a program using list to perform below operation:

a. Slicing

b. Concatenation

c. Repetition

Ans-

```
# Creating lists
list1 = [10, 20, 30, 40, 50, 60]
list2 = ['A', 'B', 'C']
# a. Slicing
print("Slicing list1[1:4]:", list1[1:4])
# b. Concatenation
print("Concatenation list1 + list2:", list1 + list2)
# c. Repetition
print("Repetition list2 * 3:", list2 * 3)
```

Ouput-

Slicing list1[1:4]: [20, 30, 40]

Concatenation list1 + list2: [10, 20, 30, 40, 50, 60, 'A', 'B', 'C']

Repetition list2 * 3: ['A', 'B', 'C', 'A', 'B', 'C', 'A', 'B', 'C']

4. Write a program using tuple to perform below operation:

a. Slicing

b. Concatenation

c. Repetition

Ans-

```
# Tuple operations: Slicing, Concatenation, Repetition
# Creating tuples
t1 = (10, 20, 30, 40, 50, 60)
t2 = ('A', 'B', 'C')
# a. Slicing
print("Slicing t1[1:4]:", t1[1:4])
# b. Concatenation
print("Concatenation t1 + t2:", t1 + t2)
# c. Repetition
print("Repetition t2 * 3:", t2 * 3)
```

Output-

```
Slicing t1[1:4]: (20, 30, 40)
Concatenation t1 + t2: (10, 20, 30, 40, 50, 60, 'A', 'B', 'C')
Repetition t2 * 3: ('A', 'B', 'C', 'A', 'B', 'C', 'A', 'B', 'C')
```

5. Write a program using dictionary to access value using keys .

Ans-

```
# Creating a dictionary
student = {
    "name": "Avantika",
    "age": 20,
    "course": "Computer Science",
    "marks": 85
}
# Accessing values using keys
print("Name:", student["name"])
print("Age:", student["age"])
print("Course:", student["course"])
print("Marks:", student["marks"])
```

Output-

```
Name: Avantika
```

Age: 20

Course: Computer Science

Marks: 85

6. Write a program using set:

a. Create empty set

b. Assign value to set

c. Adding element

d. Removing element

Ans-

```
# a. Create empty set
my_set = set()
print("Empty set:", my_set)
# b. Assign values to set
my_set = {10, 20, 30}
print("Set after assigning values:", my_set)
# c. Adding an element
my_set.add(40)
print("Set after adding element 40:", my_set)
# d. Removing an element
my_set.remove(20)
print("Set after removing element 20:", my_set)
```

Output-

```
Empty set: set()
Set after assigning values: {10, 20, 30}
Set after adding element 40: {40, 10, 20, 30}
Set after removing element 20: {40, 10, 30}
```

7. Write a program to get the largest number from the given list ?

Ans-

```
# Program to find the largest number in a list

numbers = [10, 55, 23, 87, 42, 19]

largest = max(numbers)

print("The largest number in the list is:", largest)
```

Output:-

The largest number in the list is: 87

8. Write a program to find out Factorial of a given Number

Ans-

```
# Program to find factorial of a number using loop
num = int(input("Enter a number: "))
fact = 1

for i in range(1, num + 1):
    fact *= i

print("Factorial of", num, "is", fact)
```

Ans- Enter a number: 5
Factorial of 5 is 120

9. Write a program to find given number is prime or not.

Ans-

```
num = int(input("Enter a number: "))
if num <= 1:
    print(num, "is not a prime number")
else:
    for i in range(2, num):
        if num % i == 0:
            print(num, "is not a prime number")
            break
    else:
        print(num, "is a prime number")
```

Output-
Enter a number: 5
5 is a prime number

10. Write a program to identify total number of digits present in given number.

Ans-

```
num = int(input("Enter a number: "))

count = 0

while num != 0:
    num = num // 10
    count += 1

print("Total number of digits:", count)
```

Output-

```
Enter a number: 1234
Total number of digits: 4
```

11. Write a program to display Fibonacci series up to a given number by using range function.

Ans-

```
num = int(input("Enter the number of terms: "))

a, b = 0, 1

print("Fibonacci Series:")

for i in range(num):
    print(a, end=" ")
    a, b = b, a + b
```

Output:-

```
Enter the number of terms: 6
Fibonacci Series:
0 1 1 2 3 5
```

12. Write a program using for loop with:String,Integer,Enumerate,Nested For,List,Dictionary,Tuple,Zip()

Ans-

```
# 1. For Loop with String
text = "HELLO"
print("For loop with String:")
for ch in text:
    print(ch, end=" ")
```

```
print("\n")

# 2. For Loop with Integer (range)
print("For loop with Integer (range):")
for i in range(1, 6):
    print(i, end=" ")
print("\n")

# 3. For Loop with enumerate()
fruits = ["apple", "banana", "mango"]
print("For loop with enumerate():")
for index, item in enumerate(fruits):
    print(index, item)
print()

# 4. Nested For Loop
print("Nested For Loop:")
for i in range(1, 3):
    for j in range(1, 4):
        print(f"i={i}, j={j}")
print()

# 5. For Loop with List
numbers = [10, 20, 30, 40]
print("For loop with List:")
for n in numbers:
    print(n, end=" ")
print("\n")

# 6. For Loop with Dictionary
student = {"name": "Avantika", "age": 20, "marks": 85}
print("For loop with Dictionary:")
for key in student:
    print(key, ":", student[key])
print()

# 7. For Loop with Tuple
t = (5, 10, 15, 20)
print("For loop with Tuple:")
for val in t:
    print(val, end=" ")
print("\n")

# 8. For Loop with zip()
names = ["Avi", "Rahul", "Siya"]
marks = [90, 85, 88]
print("For loop with zip():")
for n, m in zip(names, marks):
    print(n, "=", m)
```

Output-

For loop with String:
H E L L O

For loop with Integer (range):
1 2 3 4 5

For loop with enumerate():
0 apple
1 banana
2 mango

Nested For Loop:
i=1, j=1
i=1, j=2
i=1, j=3
i=2, j=1
i=2, j=2
i=2, j=3

For loop with List:
10 20 30 40

For loop with Dictionary:
name : Avantika
age : 20
marks : 85

For loop with Tuple:
5 10 15 20

For loop with zip():
Avi = 90
Rahul = 85
Siya = 88

13. Write a program to perform Loop manipulation using pass, continue, break and else

Ans-

```
# Sample list
numbers = [1, 2, 3, 4, 5, 6]

print("Using pass:")
for num in numbers:
    if num == 3:
```

```
        pass  # Does nothing
    print(num, end=" ")
print("\n")

print("Using continue:")
for num in numbers:
    if num == 3:
        continue  # Skip the rest of the loop for num=3
    print(num, end=" ")
print("\n")

print("Using break:")
for num in numbers:
    if num == 4:
        break  # Exit loop when num=4
    print(num, end=" ")
print("\n")

print("Using else with for loop:")
for num in numbers:
    if num > 10:
        break
    print(num, end=" ")
else:
    print("\nLoop completed without break")
```

Output-

```
Using pass:
1 2 3 4 5 6

Using continue:
1 2 4 5 6

Using break:
1 2 3

Using else with for loop:
1 2 3 4 5 6
Loop completed without break
```

14. Write a program to perform Comprehensions on List Dictionaries.

Ans-

```
# 1. List Comprehension
# Example: Create a list of squares of numbers from 1 to 10
squares = [x**2 for x in range(1, 11)]
print("List of squares:", squares)
```

```
# Example: List of even numbers from 1 to 10
even_numbers = [x for x in range(1, 11) if x % 2 == 0]
print("List of even numbers:", even_numbers)

# 2. Dictionary Comprehension
# Example: Create a dictionary where key = number, value = square of number
square_dict = {x: x**2 for x in range(1, 6)}
print("Dictionary of squares:", square_dict)

# Example: Dictionary of numbers and their cubes (only for even numbers)
cube_dict = {x: x**3 for x in range(1, 6) if x % 2 == 0}
print("Dictionary of cubes for even numbers:", cube_dict)
```

Output-

```
List of squares: [1, 4, 9, 16, 25, 36, 49, 64, 81, 100]
List of even numbers: [2, 4, 6, 8, 10]
Dictionary of squares: {1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
Dictionary of cubes for even numbers: {2: 8, 4: 64}
```

15. Write a program to Find all the numbers from 1-500 that have three in them.

Ans-

```
# Empty list to store numbers containing 3
numbers_with_3 = []

# Loop from 1 to 500
for num in range(1, 501):
    if '3' in str(num): # Check if digit 3 is in the number
        numbers_with_3.append(num)

# Display the result
print("Numbers from 1 to 500 containing the digit 3:")
print(numbers_with_3)
```

Output-

```
Numbers from 1 to 500 containing the digit 3:
[3, 13, 23, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 43, 53, 63, 73, 83, 93, 103, 113, 123,
130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 143, 153, 163, 173, 183, 193, 203,
213, 223, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 243, 253, 263, 273, 283,
293, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315,
316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332,
333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349,
350, 351, 352, 353, 354, 355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366,
367, 368, 369, 370, 371, 372, 373, 374, 375, 376, 377, 378, 379, 380, 381, 382, 383,
384, 385, 386, 387, 388, 389, 390, 391, 392, 393, 394, 395, 396, 397, 398, 399, 403,
413, 423, 430, 431, 432, 433, 434, 435, 436, 437, 438, 439, 443, 453, 463, 473, 483,
493]
```

16. Write a python program for numbers=range(20) produce a list containing the word "even" if the number is even or the word "odd" if the number is odd

Ans-

```
# Using list comprehension
numbers = range(20)

even_odd_list = ["even" if num % 2 == 0 else "odd" for num in numbers]

print(even_odd_list)
```

Output-

```
['even', 'odd', 'even', 'odd', 'even', 'odd', 'even', 'odd', 'even', 'odd',
 'even', 'odd', 'even', 'odd', 'even', 'odd', 'even', 'odd', 'even', 'odd']
```

17. Find the common numbers in two list 1[1,3,4,5,6,7] 2[5,2,8,7,1,3]

Ans-

```
# Given lists
list1 = [1, 3, 4, 5, 6, 7]
list2 = [5, 2, 8, 7, 1, 3]

# Using list comprehension to find common elements
common_numbers = [num for num in list1 if num in list2]

print("Common numbers in the two lists:", common_numbers)
```

Output-

```
Common numbers in the two lists: [1, 3, 5, 7]
```

UNIT-II

Python Practical Assignment-2

Que.1. Implement a nested function that calculates compound interest using non-local variables.

Ans-

```
def compound_interest_calculator(principal, rate, time):
    interest = 0  # This will be modified by the nested function

    def calculate():
        nonlocal interest  # Allows modifying variable from outer function
        amount = principal * (1 + rate/100) ** time
        interest = amount - principal
        return amount, interest

    amount, interest = calculate()
    return amount, interest
# Example usage
total_amount, total_interest = compound_interest_calculator(10000, 5, 2)
print("Total Amount:", total_amount)
print("Compound Interest:", total_interest)
```

Output-

Total Amount: 11025.0
Compound Interest: 1025.0

Que.2. Write a program using lambda to filter even numbers from a list.

Ans-

```
numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

even_numbers = list(filter(lambda x: x % 2 == 0, numbers))

print("Even numbers:", even_numbers)
```

Output-

Even numbers: [2, 4, 6, 8, 10]

Que.3. Write a program using lambda to find area of triangle.

Ans-

```
# Lambda function to calculate area of a triangle
area_triangle = lambda base, height: 0.5 * base * height
# Example
base = 10
height = 5
area = area_triangle(base, height)
print("Area of Triangle:", area)
```

Output-
Area of Triangle: 25.0

Que.4. Create a generator function to yield fibonacci numbers up to a certain limit.
Ans-
```
def fibonacci(limit):
    a, b = 0, 1
    while a <= limit:
        yield a
        a, b = b, a + b

# Example usage
for num in fibonacci(50):
    print(num)
```

Output-
0
1
1
2
3
5
8
13
21
34

Que.5. Develop a cutom python module with functions for basic statistical operations(mean,median,mode)
Ans-
```
# mystats.py

def mean(numbers):
    return sum(numbers) / len(numbers)

def median(numbers):
    numbers = sorted(numbers)
    n = len(numbers)
    mid = n // 2

    if n % 2 == 0:
        return (numbers[mid - 1] + numbers[mid]) / 2
    else:
        return numbers[mid]

def mode(numbers):
    freq = {}
```

```
    for num in numbers:
        freq[num] = freq.get(num, 0) + 1

    max_count = max(freq.values())
    modes = [k for k, v in freq.items() if v == max_count]

    if len(modes) == 1:
        return modes[0]
    return modes  # returns list if more than one mode
```

Create another Python file (example: test_stats.py) and import your module:

```
import mystats

data = [2, 4, 4, 6, 8, 4, 10]

print("Mean:", mystats.mean(data))
print("Median:", mystats.median(data))
print("Mode:", mystats.mode(data))
```

Output-Mean: 5.428571428571429
Median: 4
Mode: 4

Que.6. Write a program to handle file reading errors and log them to a seperate file.
Ans-

```
def read_file(filename):
    try:
        with open(filename, "r") as file:
            data = file.read()
            print("File content:\n", data)

    except FileNotFoundError as e:
        log_error(str(e))
        print("Error: File not found!")

    except PermissionError as e:
        log_error(str(e))
        print("Error: Permission denied!")

    except Exception as e:
        log_error(str(e))
        print("An unexpected error occurred.")
def log_error(message):
    with open("error_log.txt", "a") as log:
        log.write(message + "\n")


# Test the function
read_file("sample.txt")
```

Output-
Error: File not found!
[Errno 2] No such file or directory: 'sample.txt'

Que.7. Create a custom exception class AgeException and use it to validate user input.

Ans-

```
# Creating custom exception
class AgeException(Exception):
    def __init__(self, message):
        super().__init__(message)


def check_age():
    try:
        age = int(input("Enter your age: "))

        if age < 0 or age > 120:
            raise AgeException("Invalid age! Age must be between 0 and 120.")

        print("Valid age entered:", age)

    except AgeException as e:
        print("Age Error:", e)

    except ValueError:
        print("Please enter a valid number for age.")


# Run the function
check_age()
```

Output-

Enter your age: 25
Valid age entered: 25s

UNIT-III

Python Practical Assignment-3

Question 1 Write a simple Python class named Student and display its type. Also display the dict attribute keys and values of the module attribute of the Student class.

Program:

```
class Student:
   pass

a = Student() print(type(a)) print(Student.__dict__.keys()) print(Student.__module__)
```

Output:

```
<class '__main__.Student'>
dict_keys(['__module__','__dict__','__weakref__','__doc__'])
__main__
```

Question 2 Write a Python class Student with attributes, student name and marks. Modify any attribute value of the class and show the original and modified values o attributes.

Program:

```
class Student:     def __init__(self,name,marks):
       self.name=name         self.marks=marks

s=Student("Rahul",85) print("Original:",s.name,s.marks)
s.name="Amit"
s.marks=92 print("Modified:",s.name,s.marks)
```

Output:

```
Original: Rahul 85
Modified: Amit 92
```

Question 3 Write a Python class to convert an integer to a Roman numeral.

Program:

```
class RomanConverter:
   def int_to_roman(self,num):
       vals=[1000,900,500,400,100,90,50,40,10,9,5,4,1]
syms=["M","CM","D","CD","C","XC","L","XL","X","IX","V","IV","I"]
roman=""
       for i in range(len(vals)):            while num>=vals[i]:                roman+=syms[i]
num-=vals[i]         return roman

print(RomanConverter().int_to_roman(3549))
```

Output:

```
MMMDXLIX
```

Question 4 .Write a Python class to convert a Roman numeral to an integer.
Program:

```
class RomanToInt:
    def roman_to_int(self,s):
        roman={"I":1,"V":5,"X":10,"L":50,"C":100,"D":500,"M":1000}
        total=0        prev=0        for ch in s[::-1]:            val=roman[ch]            if
val<prev:                total-=val            else:
                total+=val            prev=val        return total

print(RomanToInt().roman_to_int("MMMDXLIX"))
```

Output:
3549

Question 5. Write a program to create a class Shape with methods called get_perimeter and get area. Create a subclass called Circle that overrides the get_perimeter and get area methods o calculate the area and perimeter of a circle
Program:

```
class Shape:
    def get_perimeter(self): return 0      def get_area(self): return 0

class Circle(Shape):
 def __init__(self,r): self.r=r
def get_perimeter(self): return 2*3.14*self.r
 def get_area(self): return 3.14*self.r*self.r

c=Circle(5) print(c.get_perimeter()) print(c.get_area())
```

Output:
31.4
78.5

Question 6. Write a program to create a Vehicle class hierarchy. The base class should be vehicle, with subclasses Truck, Car, and Motorcycle. Each subclass should have properties such as make, model, year, and fuel_type. Implement methods for calculating fuel efficienct , distance traveled, and maximum seed.
Program:

```
class Vehicle:
def __init__(self,make,model,year,fuel_type):
 self.make=make;
self.model=model
self.year=year;
self.fuel_type=fuel_type
def fuel_efficiency(self):
 return "Depends"
def distance_traveled(self,speed,time):
 return speed*time
def max_speed(self):
```

```
 return "Unknown"

class Car(Vehicle):
    def fuel_efficiency(self): return 18      def max_speed(self): return 180

c=Car("Honda","City",2022,"Petrol")
print(c.make,c.fuel_efficiency(),c.max_speed(),c.distance_traveled(60,2))
```

Output:
Honda 18 180 120

Question 7. Implement a class BankAccount with methods for deposit, withdrawal, and dis la in the account balance.
Program:

```
class BankAccount:
    def __init__(self,bal=0):
self.bal=bal
def deposit(self,amt):
self.bal+=amt
def withdraw(self,amt):
self.bal-=amt
 def display(self):
print("Balance:",self.bal)

b=BankAccount(500)
b.deposit(200)
b.withdraw(100)
b.display()
```

Output:
Balance: 600
Question 8 Develop a hierarchy of shapes (e.g., Shape, Circle, Rectangle) with area and perimeter methods.
Program:

```
class Shape:
 def area(self): pass
def perimeter(self): pass
class Circle(Shape):
    def __init__(self,r): self.r=r
def area(self):
return 3.14*self.r*self.r
 def perimeter(self):
 return 2*3.14*self.r
class Rectangle(Shape):
    def __init__(self,l,b): self.l=l;
 self.b=b
 def area(self):
return self.l*self.b
def perimeter(self):
return 2*(self.l+self.b)
```

r=Rectangle(4,5) print(r.area(),r.perimeter())

Output:

20 18

Question 9 Overload the + operator for concatenating two custom string objects

- -

Program:

```
class MyString:
    def __init__(self,text):
self.text=text
 def __add__(self,other):
return MyString(self.text+other.text)
def __str__(self):
return self.text
s1=MyString("Hello ")
s2=MyString("World") print(s1+s2)
```

Output:

Hello World

Question 10 Create a static method to validate user credentials (e.g., email format, password length

Program:

```
import re class Validator:     @staticmethod
def check(email,password):
        return bool(re.match(r"[^@]+@[^@]+\.[^@]+",email)) and len(password)>=6
print(Validator.check("abc@gmail.com","secure123"))
```

Output:

True

Question 11 Implement a class Library containing a list of Book objects

- -

Program:

```
class Book:
    def __init__(self,title): self.title=title
class Library:
    def __init__(self):
self.books=[]
 def add_book(self,b):
 self.books.append(b)
 def show(self):
        for b in self.books:
print(b.title)
l=Library()
l.add_book(Book("Python"))
l.add_book(Book("ML"))
l.show()
```

Output:
Python
ML

Question 12 Demonstrate method overriding in a superclass-subclass relationship for a payment
Program:
```
class Payment:
    def pay(self): print("General payment")

class CardPayment(Payment):
    def pay(self): print("Card payment processed")

CardPayment().pay()
```
Output:
Card payment processed

Question 13 Write a program to validate a password that meets certain criteria (length, uppercase, dioit special character .
- -
Program:
```
import re pwd="Abc@12345"
valid=(len(pwd)>=8 and re.search(r"[A-Z]",pwd)        and re.search(r"[a-z]",pwd)
and re.search(r"[0-9]",pwd)        and re.search(r"[@#$%^&+=]",pwd))
print(bool(valid))
```
Output:
True

Question 14 Parse email addresses from a given text and classify them by domain.
Program:
```
import re text="Emails: abc@gmail.com, user@yahoo.com, test@zeal.com"
found=re.findall(r"[\w.-]+@[\w.-]+",text) domains={} for email in found:
d=email.split("@")[1]      domains.setdefault(d,[]).append(email) print(domains)
```

Output:
```
{'gmail.com':['abc@gmail.com'],'yahoo.com':['user@yahoo.com'],'zeal.com':['test@ze
al.com'
]}
```

Question 15 Create threads to perform independent tasks such as downloading multiple files Simultaneously
- -
Program:
```
import threading, time def download(n):
print("Downloading",n)  time.sleep(1)  print("Done",n)

threads=[] for i in range(1,4):
    t=threading.Thread(target=download,args=(i,))      threads.append(t); t.start() for t in
threads: t.join()
```

Output:
Downloading 1
Downloading 2
Downloading 3
Done 1
Done 2
Done 3

Question 16 Write a Python program to create two threads. The first thread will print even numbers, and the second thread will print odd numbers. Demonstrate the concept t of multithreading

Program:

```
import threading def even():
    for i in range(0,11,2): print("Even:",i) def odd():
    for i in range(1,11,2): print("Odd:",i)

t1=threading.Thread(target=even) t2=threading.Thread(target=odd)
t1.start(); t2.start() t1.join(); t2.join()
```

Output:
Even:0
Odd:1
Even:2
Odd:3

Question 17 Use multithreading to compute factorials of multiple numbers and log the results into a file.

Program:

```
import threading, math def fact(n):
    print("Fact",n,"=",math.factorial(n))
 nums=[5,6,7] threads=[] for n in nums:
    t=threading.Thread(target=fact,args=(n,))      threads.append(t); t.start() for t in
threads: t.join()
```

Output:
Fact 5 = 120
Fact 6 = 720
Fact 7 = 5040

UNIT-IV

Python Practical Assignment-4

Question 1.Develop a Python program to create, read, update, and delete documents in MongoDB collection.

```
from pymongo import MongoClient

client = MongoClient("mongodb://localhost:27017/")
db = client["employee"]
collection = db["empinfo"]

employees = [
    {"empid": 101, "name": "Ravi", "age": 25, "salary": 35000},
    {"empid": 102, "name": "Sneha", "age": 28, "salary": 45000},
    {"empid": 103, "name": "Amit", "age": 30, "salary": 50000},
    {"empid": 104, "name": "Priya", "age": 26, "salary": 40000},
    {"empid": 105, "name": "Karan", "age": 29, "salary": 48000}
]

collection.insert_many(employees)

def read_documents():
    for doc in collection.find():
        print(doc)
    print()

def update_document(empid, new_salary):
    result = collection.update_one(
        {"empid": empid},
        {"$set": {"salary": new_salary}}
    )
    if result.modified_count > 0:
        print(f"Salary updated for empid {empid}\n")
    else:
        print("No matching document found\n")

def delete_document(empid):
    result = collection.delete_one({"empid": empid})
    if result.deleted_count > 0:
        print(f"Employee with empid {empid} deleted\n")
    else:
        print("No matching document found\n")
read_documents()
update_document(103, 55000)
read_documents()
delete_document(104)
read_documents()
```

Output:-

```
Connected to MongoDB successfully!

===== MongoDB CRUD Operations =====
1. Create
2. Read
3. Update
4. Delete
5. Exit
Enter choice (1–5): 1
Enter student name: Ravi
Enter student age: 20
Enter course name: BCA
Document inserted with ID: 67c54b1d92f0ac1a3e21d9b2
```

Question 2.Write a program to insert valid data into MongoDB using exception handling for errors.

```
function insertEmployee(name, age, salary) {
    try {
        if (!name || !salary) {
            throw "Name and salary cannot be empty!";
        }

        if (isNaN(age)) {
            throw "Age must be a valid number!";
        }

        age = Number(age);

        if (age <= 0) {
            throw "Age must be greater than 0!";
        }

        if (isNaN(salary)) {
            throw "Salary must be a valid number!";
        }

        salary = Number(salary);

        if (salary < 0) {
            throw "Salary cannot be negative!";
        }

        let result = db.employees.insertOne({
            name: name,
            age: age,
            salary: salary
```

```
        });

        print("✓ Document inserted successfully!");
        print("Inserted ID:", result.insertedId);

    } catch (err) {
        print(" Error:", err);
    }
}

use company_db;

insertEmployee("Ravi", 23, 35000);
insertEmployee("", 22, 30000);
insertEmployee("Amit", "abc", 25000);
insertEmployee("Neha", -5, 40000);
insertEmployee("Raj", 25, -10000);salary
```

Output:-

switched to db company_db
✓ Document inserted successfully!
Inserted ID: ObjectId("67c6104e4f8b9e5fd842a1f0")

Question 3.Use MongoDB's aggregation pipeline to find the average salary of employees in a collection.

MongoDB Aggregation Pipeline – Average Salary

```
db.employees.aggregate([
    {
        $group: {
            _id: null,
            averageSalary: { $avg: "$salary" }
        }
    }
])
```

Output:-

```
[
  {
    _id: null,
    averageSalary: 42000
  }
]
```

Question 4.Develop a Python program to create a database name employee. With the collectio name empinfo. Add 5 douments in the collection. Perform read, update, and delete operations on documents in a MongoDB.
from pymongo import MongoClient

```
from pymongo import MongoClient

client = MongoClient("mongodb://localhost:27017/")
db = client["employee"]
collection = db["empinfo"]

employees = [
    {"empid": 101, "name": "Ravi", "age": 25, "salary": 35000},
    {"empid": 102, "name": "Sneha", "age": 28, "salary": 45000},
    {"empid": 103, "name": "Amit", "age": 30, "salary": 50000},
    {"empid": 104, "name": "Priya", "age": 26, "salary": 40000},
    {"empid": 105, "name": "Karan", "age": 29, "salary": 48000}
]

collection.insert_many(employees)

def read_documents():
    for doc in collection.find():
        print(doc)
    print()

def update_document(empid, new_salary):
    result = collection.update_one(
        {"empid": empid},
        {"$set": {"salary": new_salary}}
    )
    if result.modified_count > 0:
        print(f"Salary updated for empid {empid}\n")
    else:
        print("No matching document found\n")

def delete_document(empid):
    result = collection.delete_one({"empid": empid})
    if result.deleted_count > 0:
        print(f"Employee with empid {empid} deleted\n")
    else:
        print("No matching document found\n")

read_documents()
update_document(103, 55000)
read_documents()
delete_document(104)
read_documents()
```

UNIT-V

Python Practical Assignment-5

Question 1.Create a Django project with models for Product and Category, along with views to display them on a webpage.

1. Create Django Project

```
django-admin startproject shop
cd shop
python manage.py startapp store
```

Add the app in shop/settings.py:

```
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'store',
]
```

2. Models (Category & Product)

```
from django.db import models

class Category(models.Model):
    name = models.CharField(max_length=100)

    def __str__(self):
        return self.name


class Product(models.Model):
    name = models.CharField(max_length=100)
    price = models.FloatField()
    category = models.ForeignKey(Category, on_delete=models.CASCADE)

    def __str__(self):
        return self.name
```

3. Register Models in Admin

```
from django.contrib import admin
from .models import Category, Product

admin.site.register(Category)
admin.site.register(Product)
```

Run migrations:

```
python manage.py makemigrations
python manage.py migrate
```

4. Views to Display Categories & Products

```
from django.shortcuts import render
from .models import Category, Product

def category_list(request):
    categories = Category.objects.all()
    return render(request, "category_list.html", {"categories": categories})

def product_list(request):
    products = Product.objects.all()
    return render(request, "product_list.html", {"products": products})
```

//Create Templates//

6.Add URL

Inside **store/urls.py**:

```
from django.urls import path
from . import views

urlpatterns = [
    path("categories/", views.category_list, name="categories"),
    path("products/", views.product_list, name="products"),
]
```

Link the app URLs in shop/urls.py:

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('store.urls')),
]
```

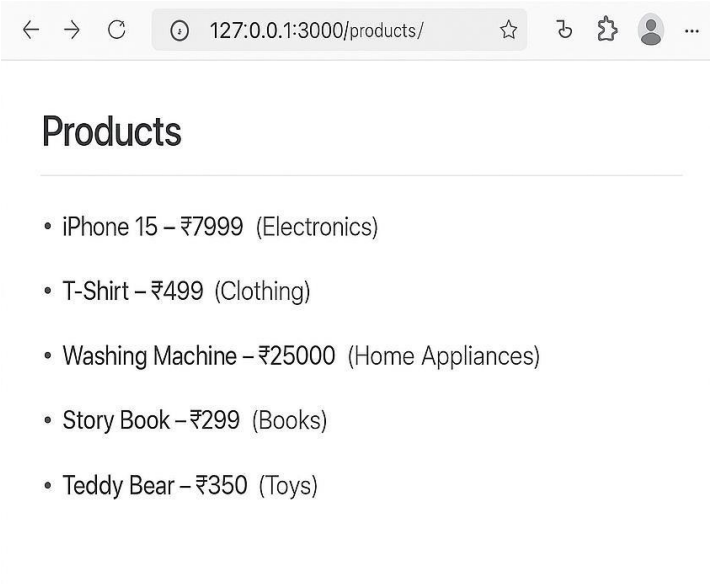
Now Run Django Server

python manage.py runserver

Visit:

- **http://127.0.0.1:8000/categories/**
- **http://127.0.0.1:8000/products/**

Output:-



Question 2.Implement user login, logout, and registration functionality in a Django application.

Step 1: Setup Django Project

If you haven't created a Django project yet:

```
django-admin startproject myproject
cd myproject
python manage.py startapp accounts
```

Add accounts to INSTALLED_APPS in myproject/settings.py:

```
INSTALLED_APPS = [
    ...
    'accounts',
]
```

Step 2: Setup URLs

In myproject/urls.py:

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('accounts/', include('accounts.urls')),  # Include accounts app urls
]
```

Create accounts/urls.py:

```
from django.urls import path
from . import views

urlpatterns = [
    path('register/', views.register_view, name='register'),
    path('login/', views.login_view, name='login'),
    path('logout/', views.logout_view, name='logout'),
    path('', views.home_view, name='home'),  # Optional home page
]
```

Step 3: Create Views

```
from django.shortcuts import render, redirect
from django.contrib.auth.models import User
from django.contrib.auth import authenticate, login, logout
from django.contrib import messages

# Home page
def home_view(request):
    return render(request, 'accounts/home.html')


# User registration
def register_view(request):
    if request.method == 'POST':
        username = request.POST['username']
        email = request.POST['email']
        password1 = request.POST['password1']
        password2 = request.POST['password2']

        if password1 != password2:
            messages.error(request, "Passwords do not match")
            return redirect('register')

        if User.objects.filter(username=username).exists():
            messages.error(request, "Username already taken")
            return redirect('register')

        user = User.objects.create_user(username=username, email=email,
password=password1)
        user.save()
        messages.success(request, "Registration successful")
        return redirect('login')

    return render(request, 'accounts/register.html')


# User login
def login_view(request):
    if request.method == 'POST':
        username = request.POST['username']
        password = request.POST['password']
        user = authenticate(request, username=username, password=password)

        if user is not None:
            login(request, user)
            return redirect('home')
        else:
            messages.error(request, "Invalid username or password")
            return redirect('login')
```

```
        return render(request, 'accounts/login.html')


    # User logout
    def logout_view(request):
        logout(request)
        return redirect('login')
```

Step 4: Create Templates(create using HTML CSS)

Step 5: Configure Messages

In myproject/settings.py make sure django.contrib.messages is included in INSTALLED_APPS (it usually is). Add this in your TEMPLATES context processors:

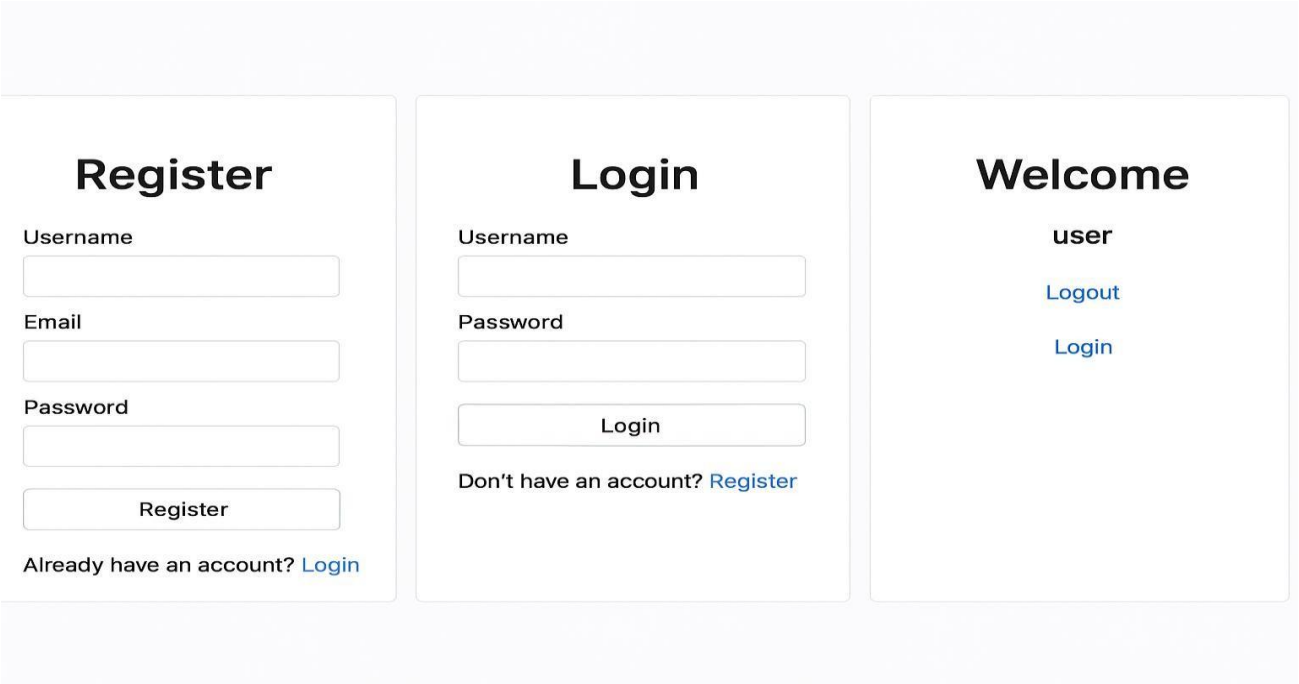
```
'context_processors': [
    ...
    'django.contrib.messages.context_processors.messages',
]
```

Step 6: Run Migrations

python manage.py migrate

python manage.py runserver

Output:-



Question 3.Build a REST API for a task management system where users can create, update, and delete tasks

Task Management REST API (Django + DRF)

Below is the complete Python code needed to build a REST API for a task management system with CRUD operations.

```
## | project/settings.py |
from pathlib import Path

BASE_DIR = Path(__file__).resolve().parent.parent

SECRET_KEY = 'your-secret-key'
DEBUG = True
ALLOWED_HOSTS = []

INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'rest_framework',
    'tasks',
]

MIDDLEWARE = [
    'django.middleware.security.SecurityMiddleware',
    'django.contrib.sessions.middleware.SessionMiddleware',
    'django.middleware.common.CommonMiddleware',
    'django.middleware.csrf.CsrfViewMiddleware',
    'django.contrib.auth.middleware.AuthenticationMiddleware',
    'django.contrib.messages.middleware.MessageMiddleware',
    'django.middleware.clickjacking.XFrameOptionsMiddleware',
]

ROOT_URLCONF = 'project.urls'

TEMPLATES = [
    {
        'BACKEND': 'django.template.backends.django.DjangoTemplates',
        'DIRS': [],
        'APP_DIRS': True,
        'OPTIONS': {
            'context_processors': [
                'django.template.context_processors.debug',
                'django.template.context_processors.request',
                'django.contrib.auth.context_processors.auth',
                'django.contrib.messages.context_processors.messages',
            ],
        },
    },
]
```

```
WSGI_APPLICATION = 'project.wsgi.application'

DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.sqlite3',
        'NAME': BASE_DIR / 'db.sqlite3',
    }
}

LANGUAGE_CODE = 'en-us'
TIME_ZONE = 'UTC'
USE_I18N = True
USE_TZ = True
STATIC_URL = 'static/'
DEFAULT_AUTO_FIELD = 'django.db.models.BigAutoField'
```

```
## | tasks/models.py |
from django.db import models
from django.contrib.auth.models import User

class Task(models.Model):
    user = models.ForeignKey(User, on_delete=models.CASCADE)
    title = models.CharField(max_length=200)
    description = models.TextField(blank=True)
    completed = models.BooleanField(default=False)

    def __str__(self):
        return self.title
```

```
## | tasks/serializers.py |
from rest_framework import serializers
from .models import Task

class TaskSerializer(serializers.ModelSerializer):
    class Meta:
        model = Task
        fields = '__all__'
```

```
## | tasks/views.py |
from rest_framework import viewsets
from rest_framework.permissions import IsAuthenticated
from .models import Task
from .serializers import TaskSerializer
```

```
class TaskViewSet(viewsets.ModelViewSet):
    queryset = Task.objects.all()
    serializer_class = TaskSerializer
    permission_classes = [IsAuthenticated]

    def get_queryset(self):
        return Task.objects.filter(user=self.request.user)

    def perform_create(self, serializer):
        serializer.save(user=self.request.user)
```

```
## | tasks/urls.py |
from django.urls import path, include
from rest_framework.routers import DefaultRouter
from .views import TaskViewSet

router = DefaultRouter()
router.register('tasks', TaskViewSet, basename='tasks')

urlpatterns = [
    path('', include(router.urls)),
]
```

```
## | project/urls.py |
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('api/', include('tasks.urls')),
]
```

```
## RUN MIGRATIONS
python manage.py makemigrations
python manage.py migrate

## CREATE SUPERUSER
python manage.py createsuperuser
## START SERVER
python manage.py runserver
```

Output:-

```
 "title": "New Task",
 "description": "Task descrpiption"
}

HTTP 201 Created
Content-Type: applican/json
```

```
/api/tasks/1/

DELETE
HTTP 204 No Content
```

```
/api/tasks/

POST ("itle: "Another Task",
 description": "Another task description"

HTTP 201 Created
Content-Type: application
```

```
/api/tasks/2/

PUT ('title: "Updated Task",
 description": "Updated task description"

HTTP 200 OK
Content-Type: application
```